

Programming Assignment 2: LTL Model Checking + Spin

Due-date: April 28 at 11:59PM (via Canvas). Submissions after the deadline will not be graded unless explicit prior permission is granted.

Homework must be individual's original work. Collaborations of any form with any students (except our TA) or other faculty members are not allowed. If you have any questions/doubts/concerns, post your questions/doubts/concerns on Piazza and/or directly ask our TA or me.

1 Problem 1: Verification

1.1 Specification

Read the Section 2 of the pdf Merz.Needham.pdf. The author discusses the use of Spin model checker for verifying the correctness of a Cryptographic Protocol (Needham-Schroeder protocol). The paper presents

- description of the protocol
- code samples for modeling the protocol
- scenario for breaking the protocol

1.2 To Do

Use the code-fragment given as part of the assignment. It follows the description and code samples presented in the paper with modifications that matches with the new syntax for PROMELA.

1. Complete the model by adding Bob's behavior in the model. Note that Alice's behavior includes a statusA flag which is set to 1, whenever Alice completes the sequence of actions (as sender) as required by the protocol. Set a similar flag in Bob's behavior (statusB) which is set to 1, whenever Bob completes the sequence of actions (as receiver) as required by the protocol.
2. Write an LTL specification, which states: whenever Alice and Bob have their status flags set to 1, they consider each other as their partner (first property in Section 2.3).
3. Run Spin model checker and produce the counter-example in terms of sends and receive operations. Copy-paste this counter-example at the top of your file.
4. Modify the protocol to correct the protocol. Explain, why the modification works.

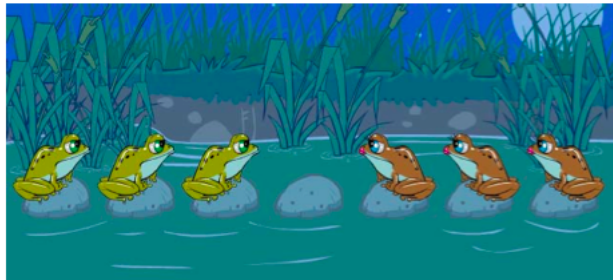
1.3 To Submit

Submit one file, PA2-P1-⟨your-net-id⟩.txt. It should contain the model of the flawed protocol along with the property that is used to prove the flaw. As part of the comments at the top of the file, explain the modification to the protocol (and the corresponding change in the Spin model) that will mitigate the flaw.

2 Problem 2: Planning

2.1 Specification

Two sets of frogs are in deep trouble.



1. For whatever reason, none of the frogs want to turn back. So they can only move forward.
2. Furthermore, they can either jump over one frog or can jump to the rock right in front of them.
3. Each rock can hold just one frog.
4. None of the frogs want to get wet.

2.2 To Do

Given the above constraints, model the movements of the frogs in Spin and use model checking an LTL specification such that you can obtain the plan to send all the right-facing frogs to the right of all the left-facing frogs.

Consider that there m frogs of each type and N pebbles where $N = 2m + 1$. Your model should be parameterized with respect to the number of pebbles N .

2.3 To Submit

Submit one file, PA2-P2-⟨your-net-id⟩.txt. At the top of the file, write the plan when $N = 7$ and $N = 11$.

COMS 512: $m = N/2$ to indicate the number of frogs of each type. For any value of m , we can record the time for model checking as presented by executing `pan -a`. Do not use any specific optimization flags of the model checker. Comment on the relationship between m and time for model checking, and discuss why such a relationship holds for this example. Write your discussion in at most 4 sentences as part of comments section in the submitted file.